Predicting Credit Card Customer Churn: Logistic Regression versus Neural Networks under Class Imbalance

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1 October 2026

Abstract

Customer churn is a central concern for credit card issuers, and identifying customers at risk of leaving is a prerequisite for any retention effort. This paper examines how accurately churn can be predicted from account and transaction data, and which features drive the prediction. Using a dataset of 10,127 credit card customers with 19 predictors and a binary target, of which 16.1 % are attrited, we compare eight model configurations: logistic regression as a baseline, a two-hidden-layer neural network, both with and without Bayesian hyperparameter optimisation, and the same four configurations after oversampling the minority class with SMOTE-NC. The best test F_1 score of 0.774 is achieved by an optimised network trained on oversampled data. Across all configurations, a single finding is stable: transaction behaviour, contact frequency and relationship depth separate the two classes, whereas sociodemographic attributes contribute almost nothing. We also document a methodological limitation that affects the level of all reported metrics: the test set serves simultaneously as the validation set during training and during hyperparameter search. We therefore treat the reported figures as upper bounds and argue that the primary obstacle in this task is not model selection but the absence of temporal information in the data.

Keywords: customer churn, credit card, class imbalance, SMOTE-NC, neural networks, hyperparameter optimisation, feature importance

1 Introduction

Customer churn, the voluntary termination of a business relationship, is a first-order concern for credit card issuers. Acquiring a new customer typically costs more than retaining an existing one, and a substantial share of the revenue generated by a card portfolio accrues not at issuance but through ongoing use: interchange fees, revolving interest, and cross-selling of additional products [15, 2]. When a customer defects, both the recurring revenue and the investment already sunk into the relationship are lost.

The practical problem faced by a bank is therefore one of identification. As the original description of the dataset used in this study puts it, the need is *"in identifying customers with high potential to leave in order to provide them with better services"* [7]. This formulation implies two distinct requirements that should not be conflated.

The first is predictive. A model must produce, for each individual customer, an assessment of how likely that customer is to defect. The output is a decision support signal answering the question of whom to approach.

The second is explanatory. The bank must know which attributes actually drive churn. Retention without that knowledge can only consist of offering the same treatment to everyone.

Only by separating relevant from irrelevant features can services be adjusted in a targeted way, which is precisely what the second half of the quoted requirement demands.

The explanatory requirement has a direct economic consequence. If churn is readable from a small set of continuously observable usage variables, then a lean data basis suffices for monitoring. If, by contrast, sociodemographic attributes were decisive, the bank would need to collect, maintain and legally safeguard sensitive personal data in order to run the same monitoring.

1.1 Research questions

This paper addresses three questions:

RQ1 How accurately can customer churn be predicted from account and transaction data, and how much does a neural network improve upon a logistic regression baseline?

RQ2 Which features carry the predictive signal, and is the resulting ranking stable across modelling choices?

RQ3 Does oversampling the minority class improve predictive performance, or does it merely shift the balance between the two error types?

1.2 Contributions

The contribution of this paper is threefold.

First, we provide a systematic comparison of eight model configurations on a single churn dataset, spanning the transition from a linear baseline to a tuned neural network and, orthogonally, the transition from the original class distribution to an oversampled one. This design isolates the effect of model capacity from the effect of class rebalancing.

Second, we show that the feature ranking is invariant across all eight configurations, whereas the level of predictive performance is not. The explanatory result is therefore considerably more robust than the predictive one.

Third, and most importantly for the interpretation of the results, we identify and document a methodological limitation that affects the level of all reported metrics. Because the test set serves simultaneously as the validation set during training and as the objective of hyperparameter search, the reported figures are optimistically biased. We quantify the consequences and treat the numbers accordingly.

1.3 Structure of the paper

Section 2 positions this work within the churn prediction literature. Section 3 describes the dataset and its quality properties. Section 4 reports the exploratory analysis that motivates the modelling hypotheses. Section 5 details the preprocessing pipeline, the evaluation framework and the eight model configurations. Section 6 presents the results. Section 7 interprets them and discusses what the results do *not* establish. Section 8 sets out the methodological limitations in detail. Section 9 concludes.

2 Related Work

Churn prediction is one of the most extensively studied applications of supervised learning in marketing and information systems. The literature can be organised along three lines that are relevant to the present study.

2.1 Predictive accuracy of churn models

Neslin et al. [15] provide the reference benchmark for churn model evaluation, arguing that predictive accuracy must be assessed with measures that account for the skewed class distribution typical of churn data, and that comparing models on overall accuracy alone is misleading. Their work established the practice of reporting measures computed specifically for the minority class, which we follow here.

Subsequent work has compared model families. Lemmens and Croux [12] show that ensemble methods based on classification trees outperform single models on churn data, while Verbeke et al. [19] demonstrate that rule-induction techniques can match the accuracy of more complex approaches while remaining interpretable. This tension between predictive performance and interpretability is directly relevant to the present study, where the explanatory requirement is part of the problem statement itself.

2.2 Class imbalance

Churn datasets are invariably imbalanced, with the defecting class typically constituting a small minority. He and Garcia [6] survey the resulting difficulties, which are twofold: standard learners are biased towards the majority class, and conventional evaluation measures lose their discriminating power. For evaluation, Saito and Rehmsmeier [17] show that precision-recall analysis is more informative than ROC analysis under severe imbalance, a recommendation we follow in the discussion.

For the modelling side, the synthetic minority over-sampling technique (SMOTE) introduced by Chawla et al. [4] generates synthetic minority observations by interpolating between neighbouring minority instances. Because the original formulation is restricted to continuous features, the same authors proposed SMOTE-NC [5], which handles datasets containing nominal attributes by generating continuous values through interpolation and nominal values by taking the majority category of the neighbourhood. We use SMOTE-NC, implemented in `imbalanced-learn` [11], as the oversampling method in this study.

2.3 Methodological pitfalls

Two strands of methodological literature inform our critical assessment of the results.

Kaufman et al. [8] formalise the notion of leakage in data mining, distinguishing between legitimate and illegitimate use of information that is unavailable at prediction time. Their framework is directly applicable to the present study, in which the test set informs both the stopping decision and the hyperparameter choice.

Cawley and Talbot [3] analyse the selection bias that arises when model selection and performance evaluation are performed on the same data. Their central argument, that model selection is itself a form of fitting and therefore requires data held apart from the final evaluation, underlies the limitations discussed in Section 8.

Finally, Ascarza [2] provides an important caveat on the interpretation of churn predictions. She shows that targeting the customers predicted to be at highest risk of churning may be ineffective, because the predictions of standard churn models confound a customer's propensity to churn with their responsiveness to a retention intervention. This distinction between prediction and causal effect is discussed in Section 7.

3 Data

3.1 Source and scope

The dataset originates from the analyttica platform and was published on Kaggle under the title *Credit Card customers* [7]. It describes a portfolio of credit card holders and contains, for

each customer, attributes from three domains: demographics, relationship data, and transaction behaviour.

The raw data comprise 10,127 observations and 23 columns. Three columns are removed prior to modelling:

- `CLIENTNUM`, a surrogate identifier carrying no substantive information. Retaining it would serve no purpose for prediction and would introduce a risk that the same customer appears in both the training and the test partition.
- two columns containing the output of a pre-computed naive Bayes classifier. These values are derived from the target variable and therefore encode the very information the model is supposed to learn. Retaining them would constitute a textbook case of target leakage [8] and would produce an apparently perfect but substantively meaningless prediction.

After removal, 10,127 observations and 20 columns remain: 19 predictors and one binary target.

3.2 Variables

Of the 19 predictors, 14 are numerical and 5 are categorical. Table 1 summarises the variables and their meaning.

Variable	**Description**
Numerical predictors (14)	
`Customer_Age`	Age of the customer in years
`Dependent_count`	Number of dependants
`Months_on_book`	Length of the customer relationship in months
`Total_Relationship_Count`	Number of products held by the customer
`Months_Inactive_12_mon`	Number of inactive months in the last twelve months
`Contacts_Count_12_mon`	Number of contacts with the bank in the last twelve months
`Credit_Limit`	Credit limit granted
`Total_Revolving_Bal`	Revolving balance on the card
`Avg_Open_To_Buy`	Average credit available for spending
`Total_Amt_Chng_Q4_Q1`	Change in transaction amount from Q4 to Q1
`Total_Trans_Amt`	Total transaction amount
`Total_Trans_Ct`	Total number of transactions
`Total_Ct_Chng_Q4_Q1`	Change in transaction count from Q4 to Q1
`Avg_Utilization_Ratio`	Average credit card utilisation ratio
Categorical predictors (5)	
`Gender`	Customer gender
`Education_Level`	Highest level of education attained
`Marital_Status`	Marital status
`Income_Category`	Income bracket
`Card_Category`	Card product (Blue, Silver, Gold, Platinum)
Target variable	
`Attrition_Flag`	Attrited customer (1) or existing customer (0)

Table 1: Variables of the cleaned dataset.

3.3 Data quality

The dataset is formally complete: all 10,127 observations contain values in all 20 columns, with no missing entries. This is convenient for modelling, as no imputation strategy has to be chosen and justified. It does not, however, imply that the data are free of problems. Three properties are material for the interpretation of the results.

First, several categorical variables contain an explicit `Unknown` category, including marital status, education level and income bracket. Formally these are complete values; substantively they represent missing information encoded as a distinct level. A model can treat this level like any other, but whether `Unknown` describes a genuine churn pattern or merely reflects inconsistent data capture cannot be determined from the data.

Second, the dataset contains no temporal information. There is neither a reference date nor any indication of when a customer defected. The target variable therefore describes a state rather than an event: the model distinguishes between customers who have already left and those who have remained. This is a severe constraint for the business question, which concerns future behaviour, and it is discussed at length in Section 8.

Third, the predictors are recorded on widely differing scales and in differing units. The credit limit ranges into the tens of thousands, while the number of dependants ranges from zero to five. For procedures based on distances or gradients this is problematic without preprocessing; Section 5.1 describes the treatment adopted.

Finally, the dataset contains no information about retention measures that may have been applied, nor about the cost of contacting a customer. Both would be required to translate predictive performance into economic value, as discussed in Section 7.

4 Exploratory Analysis

Before modelling, the dataset is examined exploratively. The objective is not an exhaustive description but the answer to a single question: which features exhibit any difference between attrited and existing customers? Only features for which the two groups differ can contribute to a separation. The findings of this section therefore generate the hypotheses that the models subsequently confirm or refute.

4.1 Target distribution

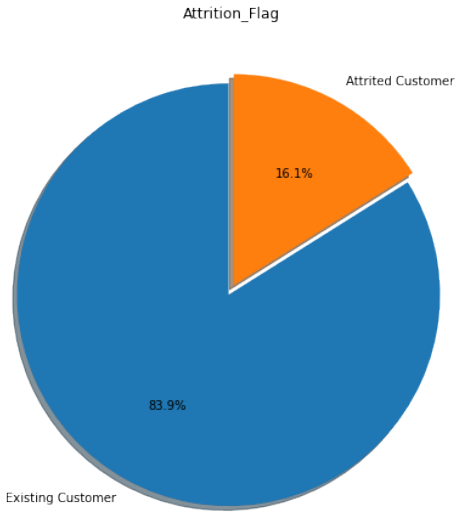


Figure 1: Share of attrited and existing customers.

Of the 10,127 customers, 16.1 % have attrited and 83.9 % have remained (Figure 1). This corresponds to approximately 1,630 attrited and 8,497 existing customers.

The resulting class imbalance has two consequences for modelling. First, the model is confronted predominantly with examples of the majority class and has little incentive to specialise

on the minority class, which is the one relevant to the business question. Second, overall accuracy is uninformative as a performance measure: a model predicting "remains" for every customer achieves 83.9 % accuracy while identifying no attrited customer at all. The imbalance is addressed in two places: through the choice of evaluation measures (Section 5.2) and through oversampling of the minority class (Section 5.7).

4.2 Categorical features

For each categorical feature, the share of attrited customers is examined by level. A level with a markedly different share indicates that the feature may be useful for discrimination.

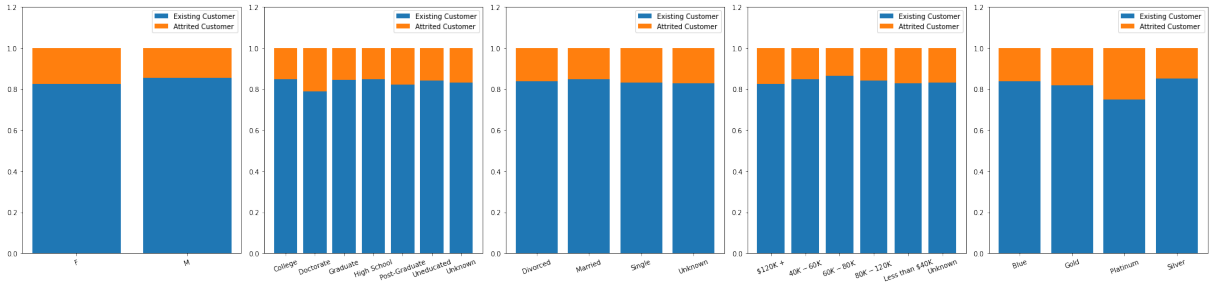


Figure 2: Share of attrited customers by level of the five categorical features.

Figure 2 reports the shares for gender, education level, marital status, income bracket and card product. The differences between levels are uniformly small, with most shares clustering around the overall rate of 16.1 %. The clearest deviation concerns the card product: holders of a Platinum card attrited more frequently, and this effect appears at both very high and very low income levels.

The finding is therefore a negative one: the categorical features separate the two groups only weakly. They are nevertheless retained in the model, since a multivariate association may exist that is hidden in a univariate view. The expectation regarding their contribution is, however, low.

4.3 Numerical features

For the numerical features, the empirical distributions are compared separately by class. If the density curves differ visibly, the feature can also contribute to discrimination on its own.

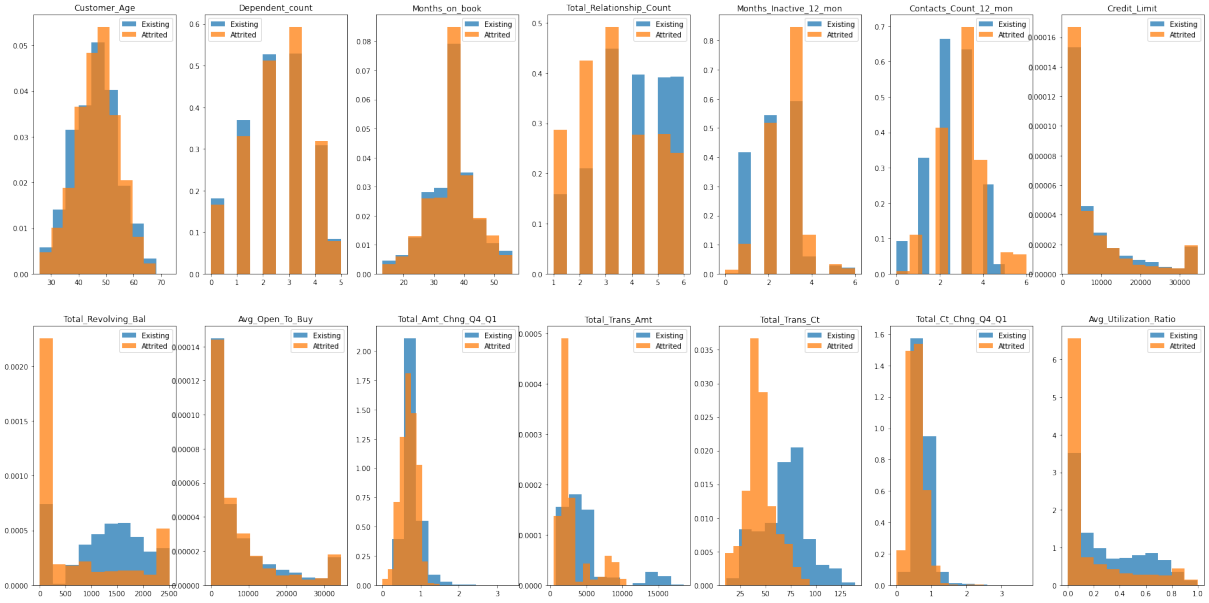


Figure 3: Empirical distributions of the 14 numerical features, by class (existing vs. attrited).

Figure 3 shows the distributions. Pronounced differences between the two classes appear for six features:

- `Total_Relationship_Count` — number of products held
- `Contacts_Count_12_mon` — contact frequency
- `Total_Revolving_Bal` — revolving balance
- `Total_Trans_Amt` — transaction amount
- `Total_Trans_Ct` — transaction count
- `Avg_Utilization_Ratio` — credit utilisation

For the remaining features — age, number of dependants, length of the relationship, credit limit and the two change ratios — the curves are largely superimposed. Notably, the sociodemographic attributes of the dataset fall almost entirely into this second group. The differences are thus concentrated in usage behaviour and in the intensity of the customer relationship, not in the person of the customer.

4.4 Outliers

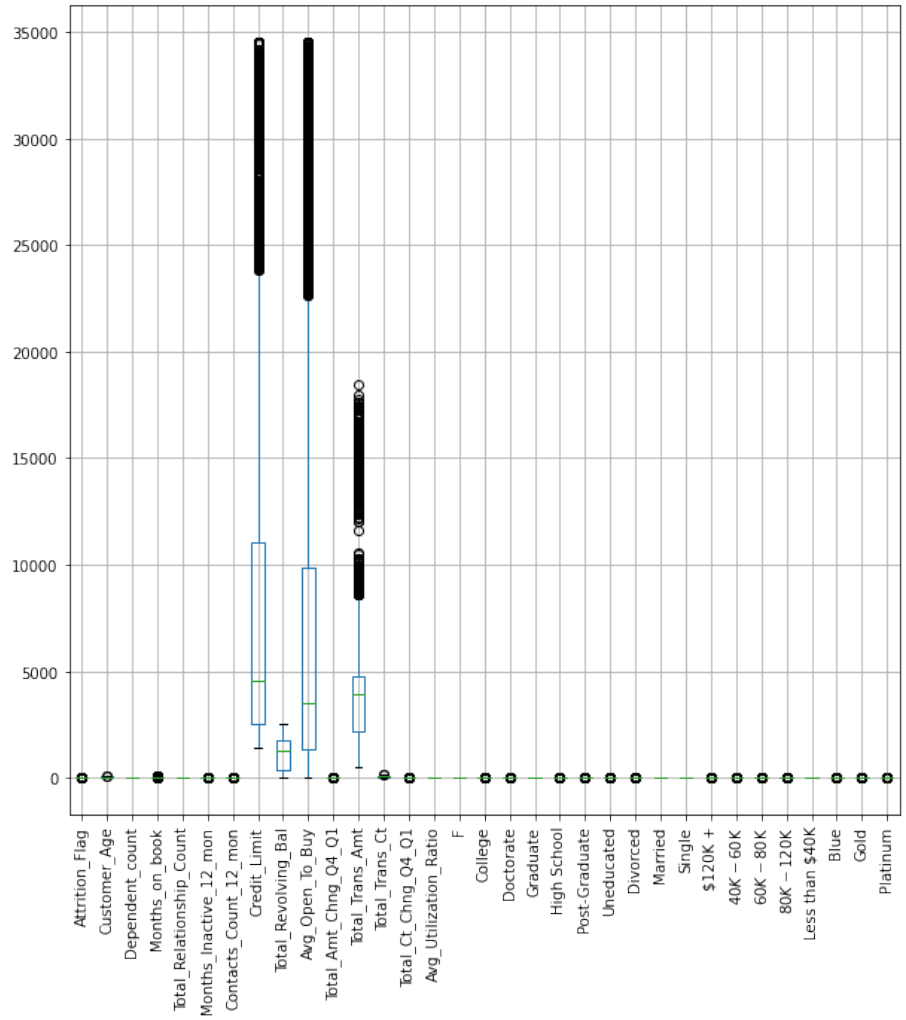


Figure 4: Boxplots of all features prior to standardisation.

Figure 4 compares the distributions across features. As expected, the features occupy very different ranges; the credit limit extends into the tens of thousands, while the number of dependants takes values between zero and five. Outliers are visible for several features.

The outliers are deliberately not removed. They are substantively plausible — a customer with a very high transaction volume is not a measurement error — and they carry information about a small customer group that may be particularly relevant to the research question. Removing them would artificially smooth the data and flatter the results. The differing scales, by contrast, are addressed through standardisation (Section 5.1).

4.5 Hypotheses

The exploratory analysis yields three expectations for the modelling stage.

H1 The separation between attrited and existing customers is achieved primarily through usage behaviour, in particular through transaction count and contact frequency.

H2 The sociodemographic attributes — gender, education, marital status, income — contribute little to the separation.

H3 No perfect result is attainable. The class-conditional distributions overlap substantially for every feature, so misclassifications are unavoidable. The question is therefore not whether errors occur, but how they are distributed between the two error types.

5 Methodology

5.1 Preprocessing

5.1.1 Encoding of categorical features

Neural networks operate on numerical inputs, so the five categorical features must be encoded. We use one-hot encoding, which creates a separate binary column for each level. For every feature, the first resulting column is dropped (the drop-first convention), since it is fully determined by the remaining columns and would contribute only redundant information.

The five categorical features thus produce 18 dummy columns. Together with the 14 numerical features, the feature space comprises 32 columns.

5.1.2 Train–test split

The dataset is split at random in a 70/30 ratio. With 10,127 observations, this yields 7,089 cases for training and 3,038 for testing. The random state is fixed at 42 to make the result reproducible.

Stratification by the target variable is not applied. At an attrition rate of 16.1 %, this is not unproblematic: the class distribution may deviate between the two partitions by chance, and the estimate of recall, which depends exclusively on the minority class, becomes correspondingly less certain. The issue is taken up in Section 8.

5.1.3 Standardisation

The features are recorded on very different scales. Because the neural network is trained via gradients, unequal scaling would distort the optimisation: features with a large range would dominate the gradient irrespective of their substantive contribution. In addition, the feature importance measure introduced in Section 5.8 would reflect the magnitude of a feature rather than its relevance.

We therefore apply standardisation, transforming each feature to zero mean and unit variance:

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j} \tag{1}$$

where $\bar{x}_j$ and s_j denote the mean and standard deviation of feature j computed on the training data.

Crucially, the scaler is fitted exclusively on the training data and then applied unchanged to the test data. Computing the mean and standard deviation over the full dataset would allow information from the test partition to enter the training process [8].

5.2 Evaluation framework

Evaluation is based on the confusion matrix and the measures derived from it. Let TP denote true positives (correctly identified attrited customers), FP false positives (customers incorrectly flagged as attrited) and FN false negatives (attrited customers classified as existing). Then

$$\text{Precision} = \frac{TP}{TP + FP} \qquad \text{Recall} = \frac{TP}{TP + FN} \tag{2}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

Precision answers how reliable a warning is: what proportion of customers flagged as at risk actually defect? Recall answers how complete the detection is: what proportion of customers who actually defect are found?

Both error types carry different costs for the bank. A false negative means a customer defects without the bank being able to react; the forgone revenue is complete. A false positive means a customer is approached who would have remained anyway; here the costs of the call, discount or special offer are incurred, but the revenue is preserved. A recall of 1 is trivially attainable by flagging every customer, at the price of offers extended to the entire customer base. A precision of 1 is attainable by flagging only the most obvious cases, at the price of missed defections. The F_1 score, the harmonic mean of both, is therefore used as the primary comparison criterion, since it weights both error types equally. This equal weighting is a choice: it is appropriate only if a missed defector and an unnecessary offer are considered roughly equally costly. A cost-weighted variant would require a cost model that the available data do not support (Section 8).

The decision threshold is fixed at 0.5 throughout: an observation is classified as attrited if the estimated probability of class 1 exceeds this value. All measures are reported separately for training and test data. Comparing the two allows a statement about overfitting: a model performing markedly better on training than on test data has learned patterns rather than generalisable structure [3].

5.3 Baseline: logistic regression

Logistic regression serves as the baseline. It is appropriate for three reasons: it is the standard procedure for binary targets, it is stably estimable with 32 features and roughly 7,000 observations, and its coefficients are directly interpretable. Owing to the preceding standardisation, the coefficients are mutually comparable — a large absolute coefficient indicates an influential feature.

The logistic regression is thus not a competing model but the benchmark against which the incremental value of the neural network is measured [15].

5.4 Neural network

The neural network is constructed as a sequential model with two hidden layers: 20 neurons in the first and 10 in the second, both using the hyperbolic tangent activation function. The output layer consists of a single neuron with sigmoid activation, producing a probability between zero and one.

Binary cross-entropy serves as the loss function and Adam [9] as the optimiser. Each hidden layer is followed by a dropout layer with a rate of 0.25 [18], which randomly deactivates a quarter of the neurons during training and thereby counteracts overfitting. This addition was made after an initial training run in which overfitting became apparent.

Training is bounded by early stopping [16]: if the validation loss fails to improve for two consecutive epochs, training is terminated. The maximum number of epochs is 50.

5.5 Hyperparameter optimisation

The network architecture is a choice, not a derivation. To examine whether a different configuration performs better, we conduct a Bayesian hyperparameter search using Optuna [1]. An objective function creates a new network for each trial with the proposed values, trains it, and returns the achieved F_1 score; Optuna selects subsequent proposals so as to improve the objective. A total of 20 trials is run. Table 2 lists the search space.

Hyperparameter	Search space
Number of hidden layers	1 to 3
Units per layer	4 to 128 (logarithmic)
Activation function	tanh, sigmoid, relu
Dropout rate	0.1 to 0.5
Optimiser	Adam, RMSprop
Early stopping patience	1 to 5
Maximum epochs	10 to 100
Number of folds	2 to 5 (cross-validation variant only)

Table 2: Search space of the hyperparameter optimisation. In the cross-validation variant without oversampling, the dropout rate was searched over the wider interval 0.0 to 1.0.

The objective function optimises the F_1 score rather than accuracy, which is consistent with the evaluation framework of Section 5.2. It has, however, a consequence that is examined in detail in Section 8: the F_1 score is computed on the test data, so the hyperparameter search is optimised directly on the data on which performance is subsequently reported.

5.6 Cross-validation variant

In a second variant, the hyperparameter search is combined with cross-validation. The training data are partitioned into k folds; the network is trained k times, with a different fold serving as the validation set each time. The number of folds is itself part of the search and varies between 2 and 5.

The advantage lies in generalisation: hyperparameters are assessed across several data partitions and adapt less strongly to any single one [10]. The disadvantage lies in data consumption: each partition leaves less training material, which lowers the attainable performance of the individual model. The original analysis notes this trade-off explicitly. A further detail is that the network weights are reset to their initial values before each fold, so that the folds are evaluated independently.

5.7 Oversampling with SMOTE-NC

The models described so far are trained predominantly on examples of the majority class. An established countermeasure is the generation of synthetic observations of the minority class. Because the dataset contains both numerical and categorical features, we use the SMOTE-NC variant [5], which is designed for mixed attribute types: new values for numerical features are interpolated, while for categorical features the majority level of the neighbourhood is chosen.

The list of categorical features comprises the 18 dummy columns at positions 14 to 31. The sampling strategy is set to 0.5: the minority class is augmented until it is half the size of the majority class, raising its share from 16.1 % to 33.3 %. The training set thereby grows from 7,089 to approximately 8,950 observations.

Because synthetic values would be generated prior to scaling if the order were reversed, the scaler is refitted after oversampling. The test data remain untouched — they are never oversampled, as this would distort the measurement of performance.

5.8 Feature importance

To rank features by their contribution, we adopt a sensitivity-based approach. For each observation, the partial derivative of the model prediction with respect to each feature is computed; it indicates how strongly the prediction shifts under a marginal change in that feature. For each feature j, these derivatives are aggregated over all n observations into a single measure ϕ_j:

$$\phi_j = \operatorname{sgn}\left(\frac{1}{n}\sum_{i=1}^{n}\frac{\partial \hat{y}_i}{\partial x_{ij}}\right) \cdot \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\frac{\partial \hat{y}_i}{\partial x_{ij}}\right)^2} \tag{4}$$

The magnitude of the root mean squared gradient measures the strength of the influence; the sign of the mean gradient indicates its direction. A positive sign means that a higher feature value increases the estimated churn probability on average. The measure is finally normalised by the sum of the absolute values of all measures, so that the values are readable as relative shares.

The advantage of this procedure is that it applies to any differentiable model without additional assumptions. Its disadvantage is that the derivative describes only local behaviour and does not capture interactions between features [14]. A SHAP-based analysis [13] would address the latter point and is identified as future work in Section 9.

6 Results

6.1 Overview

Table 3 summarises the results of all eight model configurations. Reported are precision, recall and F_1 score, each separately for training and test data.

	Precision		**Recall**		F_1	
Model	Train	Test	Train	Test	Train	Test
Logistic regression	0.7715	0.7655	0.5880	0.5464	0.6673	0.6376
Neural network (20/10)	0.8386	0.8438	0.7029	0.6754	0.7648	0.7503
Network, Optuna w/o CV	0.8907	0.8402	0.7710	0.6996	0.8265	0.7635
Network, Optuna with CV	0.8726	0.7827	0.8355	0.7480	0.8537	0.7649
SMOTE + logistic regression	0.8578	0.6420	0.8187	0.6472	0.8378	0.6446
SMOTE + neural network (20/10)	0.8816	0.6745	0.8677	0.6935	0.8746	0.6839
SMOTE + Optuna w/o CV	0.9374	0.7655	0.9208	0.7702	0.9290	0.7678
SMOTE + Optuna with CV	0.9440	0.7679	0.9281	0.7802	0.9360	0.7740

Table 3: Results of all model configurations on training and test data.

Four observations follow from the overview.

The neural network clearly outperforms logistic regression. The test F_1 score rises from 0.6376 to 0.7503, recall from 0.5464 to 0.6754. This is the largest single jump in the entire series: it results not from fine-tuning but from the change of method.

Hyperparameter optimisation yields a smaller but consistent gain. The test F_1 score rises to 0.7635 and 0.7649 respectively. The difference between the variants with and without cross-validation is minimal. The two variants differ, however, in their profile: the variant without cross-validation achieves the higher precision (0.8402 versus 0.7827), the variant with cross-validation the higher recall (0.7480 versus 0.6996).

Oversampling shifts the error profile rather than the overall quality. In all four pairs, test recall increases — for logistic regression from 0.5464 to 0.6472, for the network from 0.6754 to 0.6935, and for the optimised networks from 0.6996 to 0.7702 and from 0.7480 to 0.7802. Precision declines in return in almost every case. The F_1 score, as the compromise, therefore remains essentially unchanged: the gain in the best cross-validated variant amounts to 0.0091 (from 0.7649 to 0.7740).

Oversampling amplifies overfitting. In all four SMOTE variants, the gap between training and test values widens substantially. It is most pronounced for the fixed-architecture network: the F_1 score is 0.8746 on training but only 0.6839 on test, a gap of 0.1907. For

comparison, without oversampling this gap is 0.0145. The synthetic observations thus cause the model to learn patterns that do not carry over to real data. Notably, this particular variant performs worse in overall terms than the same network without oversampling (0.6839 versus 0.7503).

6.2 Baseline: logistic regression

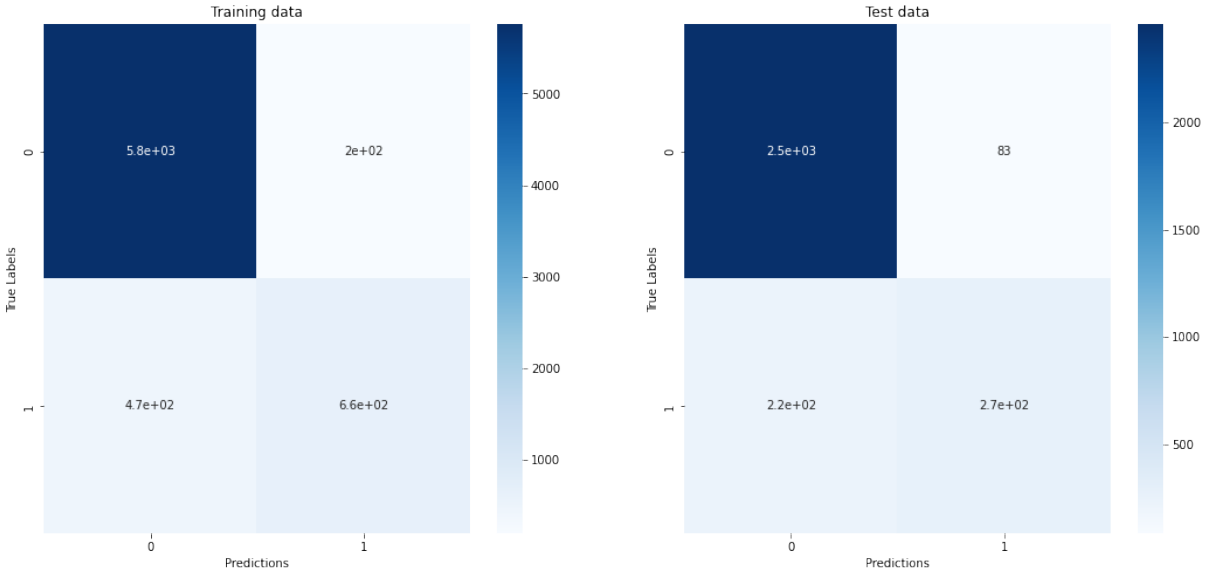


Figure 5: Confusion matrices of the logistic regression baseline.

Figure 5 shows the confusion matrices. The pattern is unambiguous: the model identifies the minority class only incompletely. A test recall of 0.5464 means that only about every second customer who actually defected was recognised as such. The precision of 0.7655 indicates in return that roughly three out of four warnings are correct.

The cause lies in the class imbalance. Since 83.9 % of training cases belong to the majority class, the model minimises overall error by predicting “remains” in case of doubt. The cost of this strategy is high in the context of the research question: missed defectors are precisely the cases the problem statement targets.

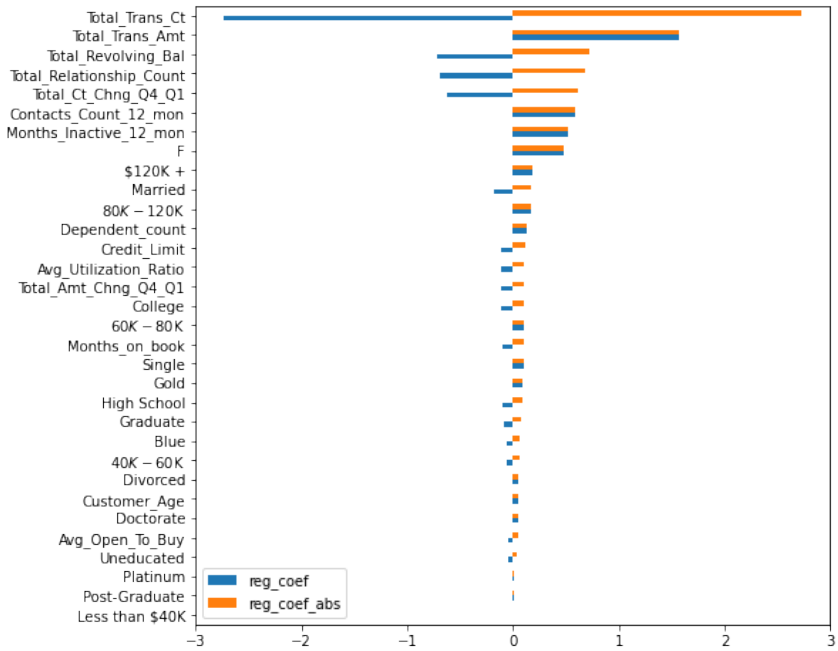


Figure 6: Regression coefficients of the logistic regression, sorted by magnitude.

Figure 6 shows the coefficients. Since the features are standardised, they are mutually comparable. The most influential features are the number of transactions, contact frequency, the revolving balance and the number of products held. The sociodemographic attributes lie at the bottom of the ranking, which corresponds exactly to the expectations from the exploratory analysis.

6.3 Neural network

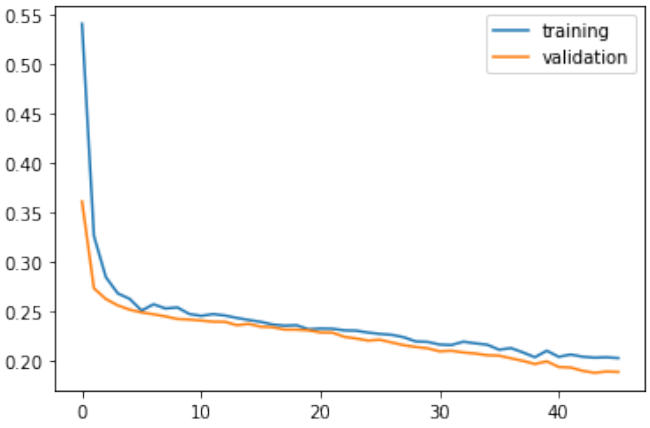


Figure 7: Loss curve over the training epochs.

Figure 7 shows the loss curve. The training loss decreases monotonically across epochs, while the validation loss falls more slowly from an early point and eventually stagnates. It was precisely this divergence that motivated the addition of dropout layers.

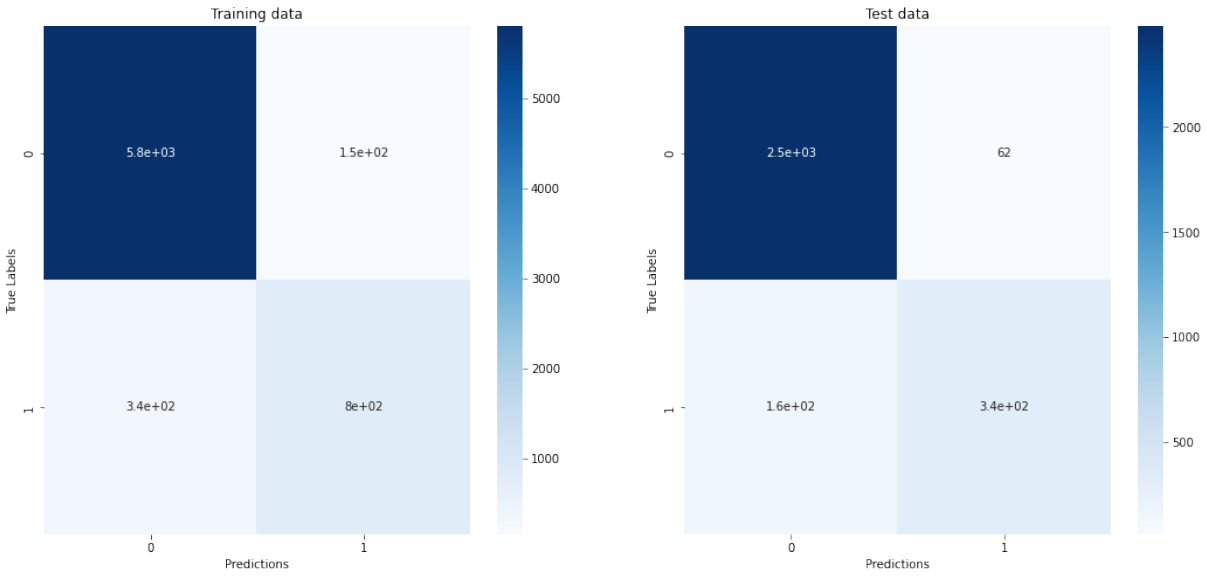


Figure 8: Confusion matrices of the neural network with two hidden layers.

Figure 8 shows the corresponding confusion matrices. Compared with logistic regression, considerably more defectors are identified without a disproportionate rise in false alarms. Recall increases by almost 13 percentage points while precision even improves slightly. The test F_1 score therefore improves by 0.1127.

This is substantively significant, as it indicates a non-linear relationship between the features and the target. Logistic regression cannot represent such relationships; the network can.

6.4 Optimised networks

Study	Architecture	Dropout	Optim.	Pat.	Epochs	Folds
Optuna w/o CV	1 layer, 42, ReLU	0.416	Adam	5	81	–
Optuna with CV	3 layers: 27 ReLU, 11 tanh, 115 tanh	0.040 / 0.011 / 0.777	Adam	4	63	4
SMOTE, w/o CV	1 layer, 127, ReLU	0.247	Adam	3	74	–
SMOTE, with CV	2 layers: 106 tanh, 32 tanh	0.287 / 0.103	Adam	3	77	5

Table 4: Best hyperparameters per optimisation study.

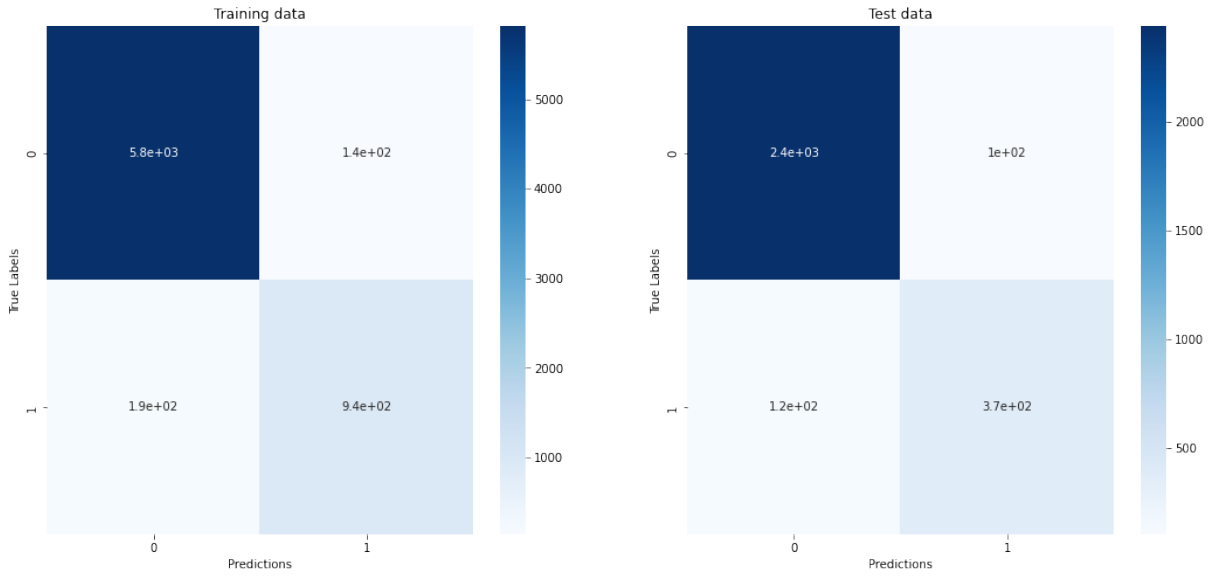


Figure 9: Confusion matrices of the optimised network with cross-validation.

Table 4 lists the configurations selected by the search and Figure 9 the corresponding confusion matrices; the matrices of the variant without cross-validation are near-identical in shape.

Notably, the search without cross-validation selects a considerably simpler architecture — a single layer with 42 neurons — and thereby achieves the higher precision. The search with cross-validation selects a larger model with three layers and achieves the higher recall instead. This suggests that the simpler architecture produces better-calibrated probabilities, while the larger model reacts more aggressively to the minority class. The available results cannot substantiate this conjecture, as no calibration analysis was performed.

It is further notable that Adam was selected as the optimiser in both studies, and that RMSprop did not prevail in any of the 40 trials. This is not evidence of Adam's superiority, but an indication that the search space is rather flat overall: the spread of achieved F_1 scores across trials is considerable — ranging from 0.0 to 0.767 and to 0.781 respectively — without a clearly preferred region emerging.

6.5 Oversampling variants

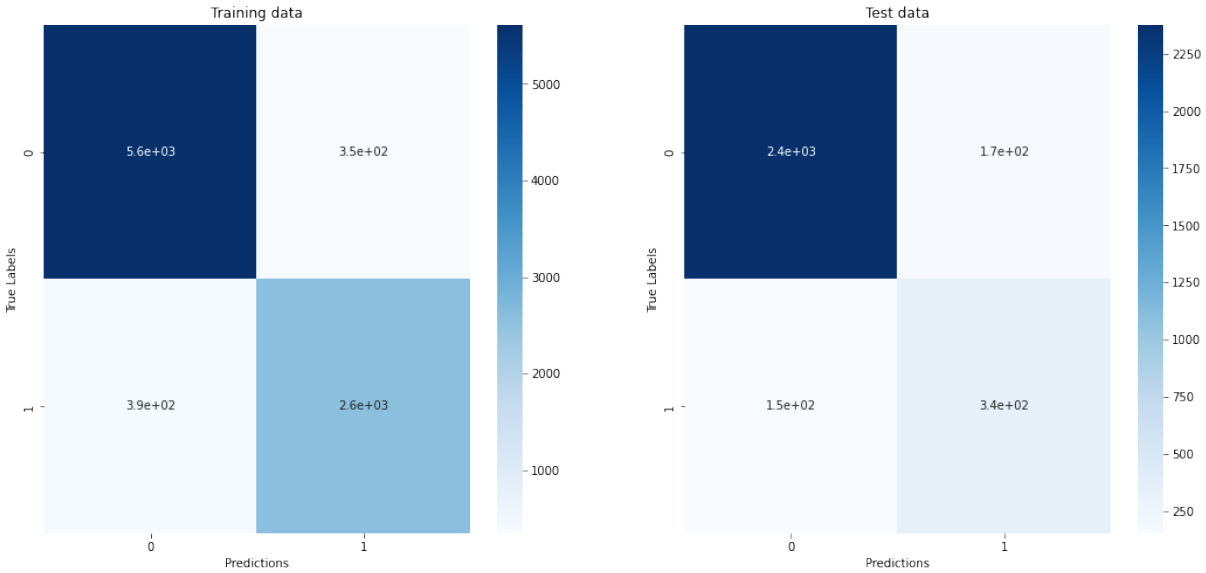


Figure 10: Confusion matrices of the neural network after oversampling.

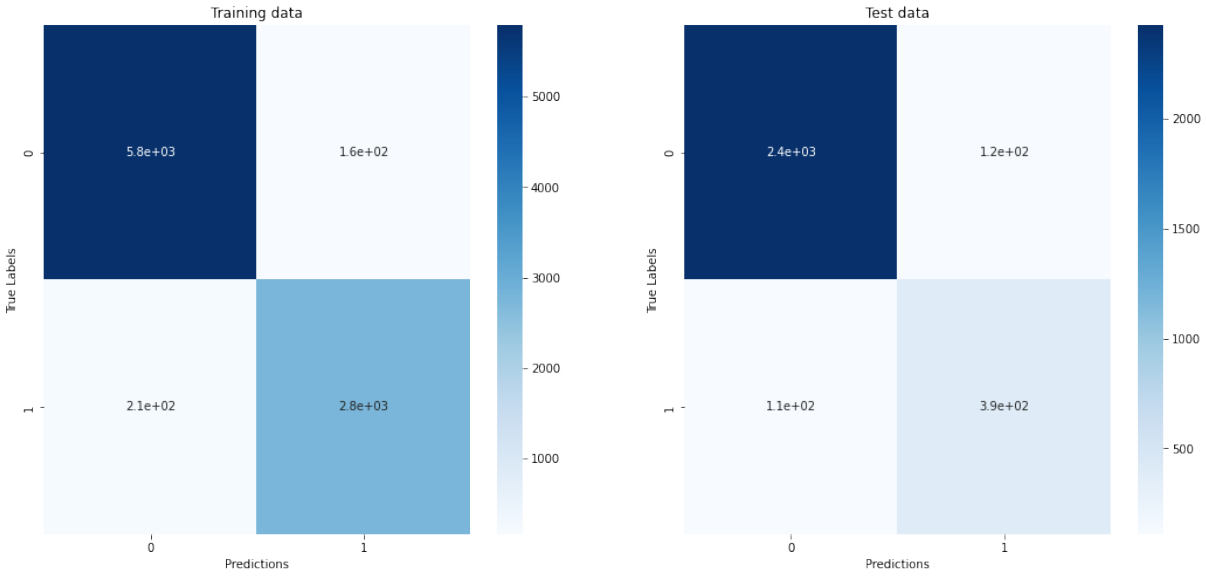


Figure 11: Confusion matrices of the optimised network with cross-validation after oversampling.

Figures 10 and 11 show the confusion matrices after oversampling. The pattern is the same in every case: the number of missed defectors falls, the number of false alarms rises. Oversampling thus shifts the decision boundary towards the minority class — precisely as intended.

The price for this is visible in two places. First, precision declines markedly in the non-optimised variants: for logistic regression from 0.7655 to 0.6420. Second, the gap between training and test values grows considerably, indicating increasing overfitting. For logistic regression, the F_1 gap is now 0.1932 rather than 0.0297.

In the overall balance, the benefit of oversampling is therefore smaller than the gain in recall suggests. The best F_1 score overall — 0.7740 — is indeed achieved with oversampling, but the lead over the same variant without oversampling is only 0.0091. Whether this difference is reliable cannot be decided without an estimate of variance (Section 8).

6.6 Feature importance

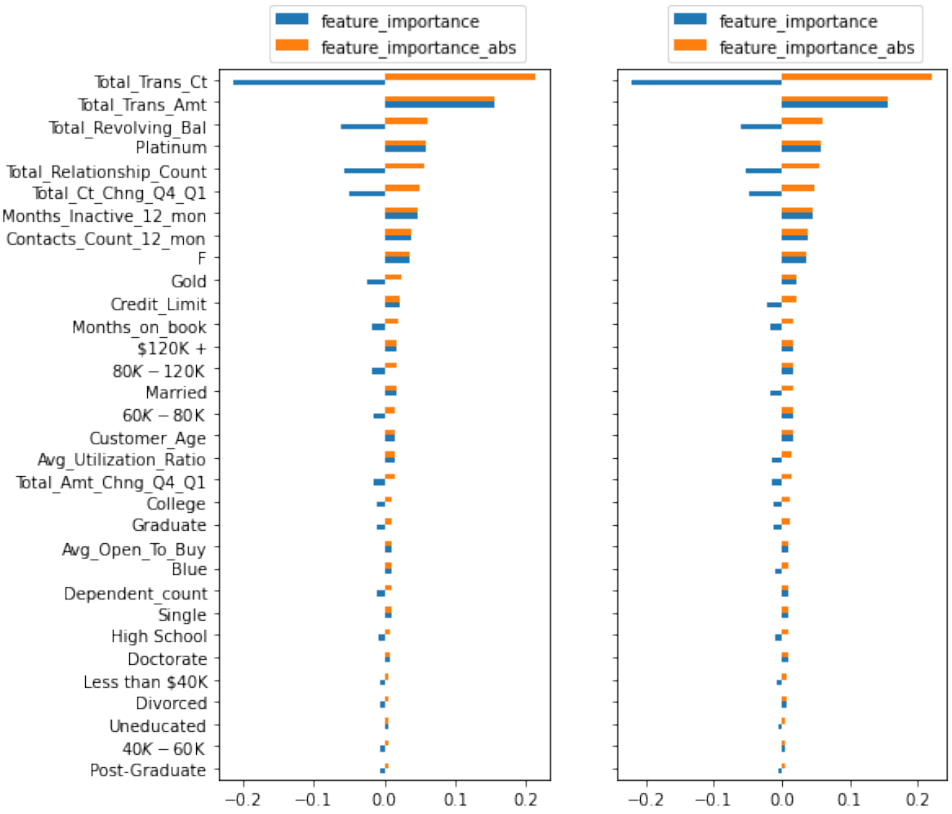


Figure 12: Feature importance of the neural network, separately for training and test data.

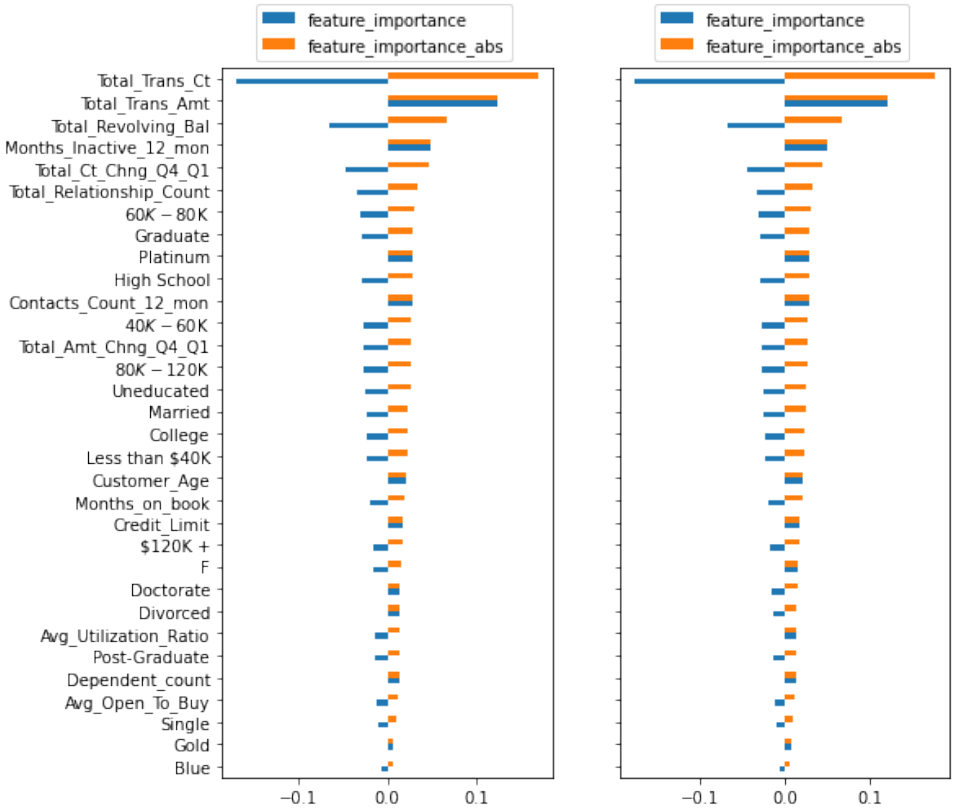


Figure 13: Feature importance of the optimised network with cross-validation after oversampling.

Figures 12 and 13 show the feature importance for two of the variants; the remaining six representations are almost identical.

The result is stable across all eight configurations. The ranking is headed by transaction count (`Total_Trans_Ct`), transaction amount (`Total_Trans_Amt`), revolving balance (`Total_Revolving_Bal`), number of products held (`Total_Relationship_Count`), contact frequency (`Contacts_Count_12_mon`), inactive months (`Months_Inactive_12_mon`) and credit utilisation (`Avg_Utilization_Ratio`).

The sociodemographic attributes — gender, education level, marital status and income bracket — lie consistently at the bottom of the ranking and contribute no discernible amount to prediction in any variant.

The model thus confirms hypothesis H1 and H2 from Section 4 in full. Consequently, the most important substantive finding of this study is not the attained F_1 score but the fact that churn is readable from a small number of continuously observable usage variables.

7 Discussion

7.1 The stable explanatory finding

The most robust result of this study is not a single metric but a pattern that holds across all eight model configurations: a customer's churn is readable from their usage behaviour, not from their person.

Specifically, seven features consistently occupy the top of the ranking: transaction count, transaction amount, revolving balance, number of products held, contact frequency, number of inactive months, and credit utilisation. Gender, education, marital status and income contribute essentially nothing to the separation. That this ranking survives both the change of model family and the change of class distribution — the two most substantial manipulations in the experimental design — is what makes it credible.

For the bank this has an immediate consequence: monitoring churn risk requires no sociodemographic data. It suffices to evaluate continuously the indicators that accrue in the transaction system anyway. This reduces the effort of data collection and maintenance and, at the same time, avoids the handling of particularly sensitive personal data. That the explanatory requirement of the problem statement can be met with a lean data basis is, in our view, the most transferable outcome of this study.

7.2 Model choice and the precision-recall trade-off

Model choice depends on which error type is weighted more heavily. Table 3 provides the basis for that decision.

If precision is the priority — that is, the reliability of each individual warning, for instance because every contact incurs appreciable cost — then the neural network without oversampling is the best choice. It attains the highest precision of all variants (0.8438) while its F_1 score (0.7503) lies only slightly below the optimum.

If recall is the priority — that is, the completeness of detection, for instance because a lost customer is more costly than an unnecessary contact — then the combination of oversampling and the optimised cross-validated network should be chosen. It attains the highest recall (0.7802).

For operational deployment, the simplest way to trade the two objectives against each other is not the choice of model but a shift of the decision threshold. All models output a probability; the value 0.5 is merely a convention. Lowering the threshold raises recall and lowers precision, continuously. This allows the trade-off to be resolved more finely than by choosing between two model variants, and it does not require retraining. The present analysis did not exploit this possibility; we regard threshold optimisation guided by a cost model as the single most valuable extension of this work, and we note that under class imbalance precision-recall analysis is the appropriate diagnostic for it [17].

7.3 Prediction is not causation

Three statements cannot be derived from the results, although they suggest themselves.

The features are not causes. That transaction count exhibits the strongest association with churn does not mean that infrequent transactions cause churn. Both quantities may depend on a common third factor, such as a weakening attachment to the institution. A model built on such an association is useful for prediction but not automatically for intervention. A measure that merely raises the transaction count improves the metric, not necessarily the relationship.

Targeting the highest-risk customers may be ineffective. Ascarza [2] shows that standard churn models conflate a customer's propensity to churn with their responsiveness to a retention measure. Customers identified as high-risk may be precisely those who would leave regardless of any offer. Since the present dataset contains no information on retention measures or their outcomes, this distinction cannot be examined here — but it qualifies any operational use of the ranking.

The direction of the effect is only partly established. The measure in Section 5.8 includes a sign indicating the direction of influence. The original analysis, however, formulates its recommendation as monitoring customers with a *low* transaction count "in combination with higher transactions". This formulation is internally contradictory. Before operational use, the actual direction of the association must be established — a straightforward task, since the signed importance values already contain the required information.

7.4 Practical implications

Subject to the limitations set out in the next section, the following implications follow for practice.

1. **Monitor usage, not demographics.** A churn monitoring system can be built on transaction and relationship indicators alone. This is cheaper, faster and less privacy-intrusive than an approach relying on sociodemographic data.

2. **Choose the model according to the cost structure.** The neural network without oversampling is preferable when false alarms are expensive; the oversampled optimised network when missed defectors are expensive.

3. **Prioritise threshold optimisation over further model tuning.** The differences between the tuned variants are marginal (0.0091 in F_1), whereas the threshold governs the entire precision-recall trade-off.

4. **Do not read the reported metrics as expectations.** They are upper bounds, for the reasons set out below.

8 Limitations

The reported results are subject to constraints that we consider material enough to state explicitly. Several of them affect the level of every metric in Table 3; the reported figures are therefore upper bounds rather than expectations of operational performance.

8.1 The test set is not a test set

The most consequential limitation concerns the role of the data partition labelled "test". In the reported pipeline, this partition is not held back until the end of the analysis. It serves as the validation set during training — early stopping monitors its loss — and, in the optimisation studies, as the basis on which the hyperparameter search evaluates its objective. The model is thus, at every stage, selected with knowledge of the data on which its performance is subsequently reported.

The consequence is a systematic overestimation of generalisation performance. Early stopping on the test set selects the epoch that happens to fit that particular partition best; the hyperparameter search selects the configuration that does so most strongly. Neither selection criterion reflects performance on unseen data. This is the classical form of selection bias described by Cawley and Talbot [3], and it applies here in its full severity.

A methodologically clean design would reserve a third partition: training, validation for early stopping and hyperparameter selection, and a final test set touched exactly once. Our results should therefore be read as a comparative ranking under identical conditions — which they support well, since all eight configurations are affected equally — but not as estimates of out-of-sample performance.

8.2 Single split without stratification

The train–test split is a single random partition with a fixed seed, without stratification by the target variable. Two consequences follow.

First, the estimate of every metric rests on one partition only. There is no variance estimate, and no confidence interval. This matters in particular for the comparison between the best oversampled variant ($F_1 = 0.7740$) and the same variant without oversampling ($F_1 = 0.7649$). A difference of 0.0091 is well within the range that a different random split could plausibly produce, so that the superiority of oversampling cannot be regarded as established. Repeated k-fold cross-validation over the entire dataset, or at minimum several seeds, would be required to substantiate it.

Second, without stratification the class proportions in the two partitions may deviate from the population rate of 16.1 %. Since recall depends exclusively on the minority class, a slightly

larger or smaller number of attrited customers in the test partition shifts that metric directly. The effect is likely modest at 3,038 test observations, but it is uncontrolled.

8.3 Cross-validation does not eliminate the problem

The cross-validation variant was intended to mitigate the selection problem, and it does reduce the dependence of the hyperparameter choice on a single partition. It does not, however, eliminate the problem, for two reasons.

The search objective is still computed on the test partition rather than on the cross-validation folds; the folds enter only through the training procedure. The test data therefore remain part of the selection criterion. In addition, the original analysis notes that the network weights are reset before each fold and that the loop returns the performance of the final fold. Under that reading, the reported figure describes one fold rather than the average across folds, which would make the measure noisier than a genuine cross-validation estimate and would reintroduce the dependence on a single partition — now inside the training data.

We report the variant as implemented, since the comparison between variants remains informative under identical conditions, but we do not interpret its slightly higher F_1 score as evidence that cross-validation improved the model.

8.4 Oversampling and feature importance

Two further points concern the oversampling variants specifically.

SMOTE-NC generates synthetic observations by interpolating between existing minority-class cases. The synthetic points are plausible under the model of the data that the procedure assumes, but they are not observations of real customers. Where the minority class is heterogeneous — and the exploratory analysis suggests considerable spread — interpolation between distant neighbours may produce cases that lie in regions the true data do not occupy. The measured recall after oversampling is therefore, in part, a statement about the synthetic data and only indirectly about customers. The pronounced widening of the train–test gap in all four oversampled variants (Section 6) is consistent with this reading.

The sensitivity-based feature importance measure is computed as a local derivative and therefore captures only marginal effects. Interactions between features are not represented; if, for instance, the effect of contact frequency depends on the number of products held, this measure cannot show it. A game-theoretic attribution such as SHAP [13] would address this. Because the ranking proved stable across all eight configurations, we consider the qualitative conclusion robust, but the relative magnitudes should not be read too precisely.

8.5 Fixed decision threshold

All reported metrics are computed at a decision threshold of 0.5. This is a convention, not a result. Since the models are trained on imbalanced data, their estimated probabilities are not calibrated to the class frequencies: the majority class is systematically favoured. A threshold of 0.5 therefore produces a conservative classifier that reports fewer positives than the class distribution would warrant.

This affects the absolute level of recall and precision, though not necessarily the ranking of models. As noted in Section 7, threshold optimisation guided by a cost model would be the most direct route to improved operational performance, and it would not require retraining.

8.6 Status classification instead of event prediction

The target variable records whether a customer has attrited, not when. The data contain no reference date, no observation window and no timestamp. The task is therefore a classification of states, not a prediction of events.

For the business question, which concerns the future, this is the central conceptual limitation. A model that separates former customers from current ones with an F_1 score of 0.774 has demonstrated that the two groups differ in the recorded attributes. It has not demonstrated that a currently active customer with the same attribute profile will defect. The latter would require longitudinal data with a defined prediction horizon, and a formulation in which the features are observed strictly before the outcome.

All operational implications in Section 7 are subject to this reservation. We regard the absence of temporal structure as a more fundamental obstacle than any of the modelling decisions discussed above.

8.7 Reproducibility and the absence of a cost model

The random seed is fixed at 42 for the data split, which makes the partitioning reproducible. The optimisation studies, however, run 20 trials each without a recorded sampler seed, and the original analysis reports no runtime environment or library versions. An exact reproduction of the reported values is therefore not guaranteed, although the qualitative findings should be stable.

Finally, the dataset contains no cost information: neither the value of a retained customer, nor the cost of a retention contact, nor the response rate of a campaign. Every statement about which error type matters more rests on a qualitative judgement. A quantitative treatment would require the corresponding business data, and would in turn permit the threshold to be set by optimisation rather than by convention.

9 Conclusion

9.1 Summary

This paper examined how customer churn in credit card portfolios can be predicted from account and transaction data, and which features carry the prediction. Using 10,127 customer records with 19 predictors and a binary target, of which 16.1 % were attrited, we compared eight model configurations: a logistic regression baseline and a neural network, each with and without Bayesian hyperparameter optimisation, and the same four configurations after oversampling the minority class with SMOTE-NC.

Three findings emerge.

The model family matters more than the tuning. The change from logistic regression to a neural network produced the largest single improvement in the series: the test F_1 score rose from 0.6376 to 0.7503, a gain of 0.1127. The subsequent hyperparameter optimisation added only 0.0132, and the best overall variant — the optimised cross-validated network on oversampled data — reached 0.7740, a further 0.0091. The returns diminish sharply. The improvement from the change of method exceeds the improvement from all subsequent tuning combined by a factor of five.

The explanatory finding is stable. Across all eight configurations, seven features consistently occupied the top of the ranking: transaction count, transaction amount, revolving balance, number of products held, contact frequency, number of inactive months, and credit utilisation. Gender, education level, marital status and income bracket contributed essentially nothing in any configuration. That this ranking survived both the change of model family and the change of class distribution is what makes it credible, and it is the result we consider most transferable.

Oversampling shifts the error profile rather than improving the model. In every pair, recall increased and precision declined. As a compromise measure, F_1 moved by at most 0.0091. The same intervention also widened the gap between training and test performance substantially, in one case from 0.0145 to 0.1907. The benefit of oversampling in this setting is thus considerably smaller than the recall gain alone suggests.

9.2 Contributions

The contribution of this work is threefold. First, it provides a systematic comparison of eight configurations under identical conditions, which allows the incremental value of each intervention to be quantified separately. Second, it documents a methodological limitation — the use of the test partition as a validation set throughout training and model selection — and traces its consequences through the reported figures; the metrics are presented as upper bounds, and the comparative ranking is presented as the reliable part of the result. Third, it establishes that the explanatory structure of churn in this dataset is carried by behavioural rather than sociodemographic variables, which has direct consequences for the data a bank would need to maintain.

9.3 Future work

Four extensions follow directly from the limitations.

A clean evaluation design. The most urgent step is a three-way partition: training, validation for early stopping and hyperparameter selection, and a test set used once. Combined with repeated stratified cross-validation, this would replace the present point estimates with estimates carrying a variance, and would allow the small differences between the tuned variants to be assessed rather than merely observed.

Threshold optimisation under a cost model. Since the differences between the tuned variants are marginal while the decision threshold governs the entire precision–recall trade-off, the largest operational gain is likely to come from threshold selection rather than further model tuning. This requires cost information — the value of a retained customer and the cost of a contact — which the present data do not provide, and which would have to be supplied by the business side.

Temporal data and event prediction. The most fundamental limitation is the absence of a time dimension. Data with a defined observation window and a prediction horizon would permit the problem to be formulated as a genuine event prediction, and would allow the question of lead time to be addressed: how far in advance of defection is a customer identifiable as at risk? This question cannot be answered with status data at all.

Interaction-aware attribution. The sensitivity measure used here captures marginal effects only. SHAP values [13] would quantify feature contributions including interactions and would test whether the qualitative ranking reported above survives a measure with different theoretical properties. Given the stability already observed, we expect the ranking to hold, but the magnitudes to shift.

Beyond these, one direction deserves mention because it concerns the objective itself. The present study, like the analysis it reproduces, optimises the accuracy of identifying customers who will leave. Ascarza [2] shows that this is not the same as identifying customers who can be retained. A model built on the latter objective would require data on past retention measures and their outcomes, and would in all likelihood produce a different ranking from the one reported here. We consider this the most valuable long-term extension, and the one furthest from what the available data can support.

References

[1] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 2623–2631, 2019.

[2] E. Ascarza. Retention futility: Targeting high-risk customers might be ineffective. *Journal of Marketing Research*, 55(1):80–98, 2018.

[3] G. C. Cawley and N. L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107, 2010.

[4] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, 2002.

[5] N. V. Chawla, A. Lazarevic, L. O. Hall, and K. W. Bowyer. SMOTEBoost: Improving prediction of the minority class in boosting. In *Knowledge Discovery in Databases: PKDD 2003*, pages 107–119. Springer, 2003.

[6] H. He and E. A. Garcia. Learning from imbalanced data. *IEEE Transactions on Knowledge and Data Engineering*, 21(9):1263–1284, 2009.

[7] S. Goyal. Credit card customers. Kaggle dataset, 2021. `https://www.kaggle.com/datasets/sakshigoyal7/credit-card-customers`. Accessed 1 October 2026.

[8] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman. Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4):1–21, 2012.

[9] D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. In *Proceedings of the 3rd International Conference on Learning Representations (ICLR)*, 2015.

[10] R. Kohavi. A study of cross-validation and bootstrap for accuracy estimation and model selection. In *Proceedings of the 14th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 1137–1143, 1995.

[11] G. Lemaître, F. Nogueira, and C. K. Aridas. Imbalanced-learn: A Python toolbox to tackle the curse of imbalanced datasets in machine learning. *Journal of Machine Learning Research*, 18(17):1–5, 2017.

[12] A. Lemmens and C. Croux. Bagging and boosting classification trees to predict churn. *Journal of Marketing Research*, 43(2):276–286, 2006.

[13] S. M. Lundberg and S.-I. Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 4765–4774, 2017.

[14] C. Molnar. *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*. 2nd edition, 2022. `https://christophm.github.io/interpretable-ml-book/`.

[15] S. A. Neslin, S. Gupta, W. Kamakura, J. Lu, and C. H. Mason. Defection detection: Measuring and understanding the predictive accuracy of customer churn models. *Journal of Marketing Research*, 43(2):204–211, 2006.

[16] L. Prechelt. Early stopping — but when? In *Neural Networks: Tricks of the Trade*, pages 55–69. Springer, 1998.

[17] T. Saito and M. Rehmsmeier. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3):e0118432, 2015.

[18] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov. Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(1):1929–1958, 2014.

[19] W. Verbeke, K. Dejaeger, D. Martens, J. Hur, and B. Baesens. New insights into churn prediction in the telecommunication sector: A profit driven data mining approach. *European Journal of Operational Research*, 218(1):211–229, 2012.